

**BUTLER COUNTY REGIONAL, OH, USA**

WMO: 725217

Lat: 39.364N

Lon: 84.525W

Elev: 634

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53855

**Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |            |            |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD<br>to 99.6% DB |            | WSF          |
|---------------|------------|------------|-------------------------------|------------|------------|------------|------------|-------------|-----------------------|-------------|-------------|-------------|--------------------------|------------|--------------|
|               |            |            | 99.6%                         |            |            | 99%        |            |             | 0.4%                  |             | 1%          |             |                          |            |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB       | DP         | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                     | PCWD       |              |
| (a)<br>1      | (b)<br>3.0 | (c)<br>9.8 | (d)<br>-4.5                   | (e)<br>4.4 | (f)<br>6.0 | (g)<br>0.5 | (h)<br>5.8 | (i)<br>12.0 | (j)<br>21.6           | (k)<br>32.0 | (l)<br>19.1 | (m)<br>30.9 | (n)<br>3.4               | (o)<br>260 | (p)<br>0.434 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>20.5            | (c)<br>92.3     | (d)<br>74.6 | (e)<br>90.1 | (f)<br>73.9 | (g)<br>87.9 | (h)<br>73.1 | (i)<br>77.8         | (j)<br>88.3 | (k)<br>76.5 | (l)<br>86.1 | (m)<br>75.2 | (n)<br>83.9 | (o)<br>6.8           | (p)<br>220 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 74.7                            | 133.1 | 82.9 | 73.3 | 126.8 | 81.1 | 72.4 | 122.9 | 80.2 | 41.6          | 88.6 | 40.2 | 86.3 | 39.0 | 83.9 | 82.9              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |          | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|----------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |          | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max      | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 18.3          | (b) 16.1 | (c) 13.2 | (e) -5.2                   | (f) 95.2 | (g) 8.7            | (h) 3.6  | (i) -11.5                                          | (j) 97.8  | (k) -16.6  | (l) 99.9  | (m) -21.5  | (n) 102.0 | (o) -27.8  | (p) 104.6 |    |    |
| (4) 18.3          | (5) 16.1 | (6) 13.2 | (7) -5.8                   | (8) 80.6 | (9) 8.4            | (10) 1.5 | (11) -11.9                                         | (12) 81.6 | (13) -16.8 | (14) 82.5 | (15) -21.6 | (16) 83.3 | (17) -27.7 | (18) 84.4 |    |    |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 54.1   | 30.2  | 33.2  | 43.2  | 54.8  | 64.3  | 72.0  | 75.4  | 74.2  | 67.2  | 55.1  | 43.7  | 34.8  |
|                                                                        | DBStd       | 18.30  | 12.14 | 11.70 | 10.85 | 9.16  | 8.04  | 5.54  | 5.18  | 4.95  | 7.19  | 8.90  | 9.60  | 10.42 |
|                                                                        | HDD50       | 2202   | 622   | 483   | 269   | 55    | 4     | 0     | 0     | 0     | 0     | 47    | 238   | 484   |
|                                                                        | HDD65       | 5064   | 1079  | 891   | 677   | 322   | 115   | 11    | 1     | 2     | 59    | 329   | 641   | 937   |
|                                                                        | CDD50       | 3702   | 8     | 12    | 60    | 199   | 448   | 660   | 786   | 749   | 515   | 205   | 48    | 12    |
|                                                                        | CDD65       | 1089   | 0     | 0     | 3     | 16    | 95    | 221   | 322   | 286   | 124   | 22    | 0     | 0     |
|                                                                        | CDH74       | 10733  | 0     | 2     | 39    | 225   | 941   | 2075  | 3158  | 2725  | 1311  | 248   | 9     | 0     |
|                                                                        | CDH80       | 3913   | 0     | 0     | 5     | 32    | 259   | 725   | 1276  | 1071  | 487   | 57    | 1     | 0     |
| (14) Wind                                                              | WSAvg       | 5.1    | 6.6   | 6.5   | 6.5   | 6.6   | 4.8   | 4.1   | 3.6   | 3.4   | 3.5   | 4.5   | 5.2   | 5.8   |
| (15) Precipitation                                                     | PrecAvg     | 41.6   | 3.0   | 2.4   | 3.5   | 4.3   | 4.8   | 4.3   | 3.9   | 3.0   | 3.1   | 3.0   | 2.9   | 3.5   |
|                                                                        | PrecMax     | 59.6   | 6.1   | 5.4   | 9.4   | 10.8  | 10.4  | 9.1   | 8.8   | 5.7   | 7.0   | 6.0   | 6.8   | 8.6   |
|                                                                        | PrecMin     | 29.6   | 1.2   | 0.8   | 1.1   | 1.4   | 0.9   | 1.4   | 1.1   | 0.9   | 0.6   | 1.0   | 0.9   | 1.1   |
|                                                                        | PrecStd     | 6.9    | 1.2   | 1.3   | 1.6   | 2.2   | 2.4   | 1.7   | 1.7   | 1.2   | 1.9   | 1.5   | 1.4   | 1.4   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 65.0  | 70.3  | 79.1  | 83.5  | 88.8  | 92.6  | 96.6  | 95.3  | 93.4  | 86.2  | 74.6  | 65.8  |
|                                                                        |             | MCWB   | 59.1  | 59.6  | 61.6  | 64.3  | 71.5  | 74.5  | 78.2  | 74.9  | 71.9  | 71.2  | 61.0  | 59.0  |
|                                                                        | 2%          | DB     | 59.9  | 63.0  | 72.4  | 80.4  | 85.5  | 89.7  | 92.3  | 91.4  | 89.7  | 80.3  | 69.9  | 60.5  |
|                                                                        |             | MCWB   | 54.9  | 54.2  | 58.8  | 63.3  | 69.4  | 73.9  | 75.8  | 74.1  | 71.9  | 66.4  | 58.4  | 55.1  |
|                                                                        | 5%          | DB     | 54.5  | 57.2  | 66.6  | 76.1  | 83.0  | 87.6  | 89.8  | 89.0  | 85.8  | 75.8  | 65.3  | 55.4  |
|                                                                        |             | MCWB   | 50.6  | 50.9  | 55.5  | 60.9  | 68.6  | 72.8  | 75.1  | 73.3  | 70.6  | 63.9  | 57.0  | 50.9  |
|                                                                        | 10%         | DB     | 49.2  | 52.0  | 62.1  | 72.3  | 80.0  | 84.4  | 87.3  | 86.1  | 81.8  | 71.9  | 60.9  | 51.5  |
|                                                                        |             | MCWB   | 44.1  | 45.7  | 53.6  | 59.5  | 67.2  | 71.1  | 74.0  | 72.1  | 68.4  | 62.0  | 53.6  | 46.9  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 60.1  | 61.5  | 63.6  | 68.5  | 74.4  | 77.9  | 80.5  | 78.4  | 76.4  | 72.8  | 64.4  | 60.9  |
|                                                                        |             | MCDB   | 63.8  | 68.6  | 76.3  | 79.2  | 83.6  | 88.1  | 90.9  | 88.9  | 87.1  | 82.8  | 70.5  | 64.7  |
|                                                                        | 2%          | WB     | 56.0  | 57.0  | 60.6  | 65.8  | 72.2  | 76.1  | 78.4  | 77.1  | 74.7  | 69.5  | 61.1  | 55.8  |
|                                                                        |             | MCDB   | 59.6  | 60.8  | 69.3  | 75.1  | 82.2  | 85.9  | 89.0  | 87.3  | 84.3  | 75.6  | 67.2  | 58.8  |
|                                                                        | 5%          | WB     | 51.2  | 51.7  | 57.9  | 63.3  | 70.5  | 74.7  | 77.1  | 75.7  | 72.9  | 66.5  | 57.8  | 51.8  |
|                                                                        |             | MCDB   | 54.1  | 56.7  | 64.3  | 71.5  | 80.0  | 84.0  | 87.0  | 84.7  | 81.6  | 73.4  | 63.8  | 55.4  |
|                                                                        | 10%         | WB     | 44.5  | 45.5  | 54.4  | 61.0  | 68.6  | 73.2  | 75.5  | 74.3  | 71.0  | 63.5  | 53.9  | 46.8  |
|                                                                        |             | MCDB   | 48.3  | 51.2  | 61.3  | 69.9  | 77.0  | 81.7  | 84.0  | 82.2  | 78.1  | 69.2  | 59.7  | 50.2  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 17.0  | 18.9  | 20.5  | 23.2  | 21.7  | 21.0  | 20.5  | 22.0  | 24.1  | 23.6  | 21.8  | 16.4  |
|                                                                        |             | MCDBR  | 20.9  | 25.8  | 28.0  | 28.8  | 25.3  | 24.7  | 22.6  | 25.6  | 28.7  | 28.9  | 28.1  | 21.2  |
|                                                                        |             | MCWBR  | 17.5  | 19.6  | 16.4  | 15.3  | 12.2  | 11.2  | 9.2   | 10.4  | 12.6  | 15.8  | 18.4  | 17.6  |
|                                                                        | 5% WB       | MCDBR  | 19.4  | 23.3  | 23.2  | 23.8  | 21.2  | 20.2  | 20.1  | 20.3  | 22.3  | 23.2  | 22.7  | 19.4  |
|                                                                        |             | MCWBR  | 17.4  | 19.5  | 15.0  | 14.5  | 11.3  | 10.3  | 9.1   | 9.4   | 10.8  | 13.5  | 16.9  | 17.5  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.307 | 0.328 | 0.355 | 0.398 | 0.441 | 0.454 | 0.459 | 0.454 | 0.402 | 0.358 | 0.329 | 0.306 |
|                                                                        | taud        |        | 2.463 | 2.370 | 2.364 | 2.256 | 2.176 | 2.189 | 2.181 | 2.178 | 2.300 | 2.424 | 2.463 | 2.478 |
|                                                                        | Ebn at Noon |        | 275   | 282   | 285   | 278   | 267   | 263   | 261   | 259   | 266   | 268   | 263   | 266   |
|                                                                        | Edh at Noon |        | 25    | 32    | 36    | 42    | 47    | 46    | 46    | 45    | 37    | 30    | 25    | 23    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 561   | 815   | 1145  | 1543  | 1740  | 1935  | 1874  | 1735  | 1426  | 1020  | 691   | 475   |
|                                                                        | RadStd      |        | 49    | 99    | 101   | 123   | 148   | 118   | 114   | 90    | 140   | 99    | 79    | 47    |

**Historical Trends**

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +223  | N/A   |
| (47) Regional (36 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +218  | +133  |

Nomenclature: See separate page